

Design and Simulation of an Embedded LED Indicator System Using Arduino UNO

Introduction

This mini project focuses on the design and simulation of a basic embedded system using Arduino UNO. The system demonstrates alternate blinking of LEDs to represent indicator-based applications.

Objective

- To understand embedded system fundamentals
- To learn digital output control
- To perform simulation using Proteus software

Tools and Technologies

Arduino UNO, Proteus ISIS Professional, Embedded Systems Concepts

Circuit Description

Two LEDs are connected to the digital output pins of Arduino through current-limiting resistors. The circuit is designed and tested in Proteus simulation environment.

Working Principle

The embedded controller activates one LED at a time with a fixed delay, creating an alternate blinking pattern. This logic represents basic timing control used in real-time systems.

Real-Time Application

The system can be used as a pedestrian crossing indicator or warning signal system where timing-based indicators are required.

Conclusion

This mini project successfully demonstrates the concept of embedded simulation and indicator control. It provides a strong foundation for understanding real-time embedded applications.